

Alexander Milevski

Front-end / Visualization / GIS Engineer

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Software engineer building complex web applications and libraries at the intersection of creativity and technology. Specialized in web cartography, computational geometry, data visualization, and graph algorithms.

Experience

Ogma @ Linkurious

Lead R&D Engineer · 2016 - present · linkurious.com

Leading R&D on Ogma, a high-performance graph visualization library used by enterprises worldwide for fraud detection, cybersecurity, and network analysis.

Designed and implemented graph layout algorithms, spatial indexing data structures, and GIS integration. Managing a team of 3 engineers, coordinating with product and marketing.

Wikimapia

Front-end Lead · 2009 - 2014 · wikimapia.org

Built the front-end for a collaborative mapping platform that served 1.2 million unique visitors daily.

Architected and developed an OpenLayers-like, fully API-independent interactive map engine from scratch. Created collaborative editing tools, built both desktop and mobile web apps, and managed an international user community.

eVision GIS

R&D / GIS Specialist · 2014 - 2016

R&D engineer and GIS specialist for industry-leading permit-to-work (EPTW) software in the oil and gas sector.

Built a standalone GIS environment with interoperability between ArcGIS and Geoserver. Applied GIS principles to non-geographical spatial viewers. Created open-source Leaflet components. Later led development of 2D spatial viewers/editors for industrial schematics under an EU research grant.

martinez

Creator & Maintainer · 2016 - present · [github.com](#)

Martinez-Rueda polygon clipping algorithm implementation. Boolean operations on polygons: intersection, union, difference, xor. Handles multipolygons, holes, and complex geometries.



Figure 1: stars

Used in production GIS applications for spatial analysis and map overlay operations.

polygon-offset

Creator & Maintainer · 2015 - present · [github.com](#)

Polygon offset (margin/padding) library. Generates inner and outer offset polygons with configurable arc segments for rounded corners.



Figure 2: stars

Essential for buffer zone calculations, collision margins, and geometric padding operations in mapping and CAD applications.

wittgenstein

Creator · 2019 · [github.com](#)

Interactive reading experience for Ludwig Wittgenstein's Tractatus Logico-Philosophicus. Hierarchical navigation through the numbered propositions.



Figure 3: stars

Exploring the structure of one of the most influential philosophical texts of the 20th century through visualization.

moscow-rings

Creator · 2017 · [github.com](#)

Visualization of distances to Moscow's main transport rings. Interactive map showing how the city's circular road network shapes urban geography.



Figure 4: stars

Data-driven exploration of Moscow's concentric urban structure through the lens of transport infrastructure.

More Libraries

Creator & Maintainer · 2014 - present · github.com

More JavaScript libraries for computational geometry, data structures, and Leaflet plugins.

- **GreinerHormann** ★ 250 — Polygon clipping, boolean ops
- **Leaflet.Path.Transform** ★ 176 — Drag/rotate/resize for Leaflet
- **splay-tree** ★ 121 — Fast splay-tree data structure
- **avl** ★ 119 — Fast AVL tree for Node and browser

Skills

Languages	TypeScript, JavaScript, HTML/CSS, SQL, some C++
Visualization	WebGL, Canvas, SVG, D3.js, Three.js, graph algorithms
GIS	Leaflet, OpenLayers, Mapbox, ArcGIS API, Geoserver, PostGIS
Front-end	React, Vue, performance optimization, large codebase architecture
Tools	Git, Vite, Webpack, Node.js, Docker, CI/CD
Languages	English (fluent), French (conversational), German (basic), Russian (native)

Education

Moscow State Institute for Electronics and Mathematics Computer Science, 2003–2008

Thesis: Edge detection in image processing using neural networks

Moscow State Institute for Electronics and Mathematics Graphic Design, 2005–2009

Krasnopresnenskaya Art School 1992–2003, First Degree Diploma

Art & Exhibitions

Trained at Krasnopresnenskaya Art School and Moscow State Institute for Graphic Design. My practice spans drawing, painting, sculpture, and book art.

- Openbach Residence, Vincennes, 2025
- Kiff & Marais, Paris, 2025
- Openbach Residence, Vincennes, 2024

- Students Art Exhibition, CHA, Moscow, 2008
- Students Art Exhibition, CHA, Moscow, 2007
- “Artist’s Book” (book-art), Modern Arts Museum, Moscow, 2005
- Non-Fiction 2004 (book-art), CHA, Moscow, 2004

Art shop: art.milevski.co